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ROLL NO - CSB24021

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from abc import ABC, abstractmethod

class LibraryItem(ABC):
    item_count = 0    # class/static counter

    def __init__(self, title, year):
        self.title = title
        self.year = year
        LibraryItem.item_count += 1    # increment counter

    @abstractmethod
    def displayInfo(self):
        pass

class Book(LibraryItem):
    def __init__(self, title, year, author="Unknown"):    # default
argument
        super().__init__(title, year)
        self.author = author

    def displayInfo(self):
        print(f"[Book] Title: {self.title}, Year: {self.year}, Author:
{self.author}")

class DVD(LibraryItem):
    def __init__(self, title, year, duration, genre="General"):    #
default argument
        super().__init__(title, year)
        self.duration = duration
        self.genre = genre

    def displayInfo(self):
        print(f"[DVD] Title: {self.title}, Year: {self.year}, Duration:
{self.duration} mins, Genre: {self.genre}")
```

```

if __name__ == "__main__":

    # Creating objects
    item1 = Book("Python Basics", 2020, "John Doe")
    item2 = DVD("Inception", 2010, 148, "Sci-Fi")
    item3 = Book("Data Science", 2022)    # using default author

    # Polymorphism (list of LibraryItem objects)
    library_items = [item1, item2, item3]

    print("----- Library Items -----")
    for item in library_items:
        item.displayInfo()    # same method, different behavior

    # Display total items using class counter
    print(f"\nTotal Library Items: {LibraryItem.item_count}")

```

OUTPUT:

```

[Running] python -u "c:\Users\prerana\OneDrive\Desktop\PYTHON\tempCodeRunnerFile.py"
----- Library Items -----
[Book] Title: Python Basics, Year: 2020, Author: John Doe
[DVD] Title: Inception, Year: 2010, Duration: 148 mins, Genre: Sci-Fi
[Book] Title: Data Science, Year: 2022, Author: Unknown

Total Library Items: 3

[Done] exited with code=0 in 0.076 seconds

```